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Metaheuristic-driven two-stage hybrid feature selection and deep learning optimization for security in IoT systems

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ABSTRACT

Intrusion detection in Internet of Things (IoT) systems poses significant challenges due to high-dimensional, heterogeneous, and rapidly evolving network traffic. To address these issues, this study proposes a novel two-stage hybrid framework aimed at enhancing the performance and scalability of intrusion detection systems. In the initial stage, the Random Forest algorithm ranks features by importance and retains the top 50%. In the second stage, five metaheuristic algorithms refine the feature space: Genetic Algorithm, Particle Swarm Optimization, Whale Optimization Algorithm, Sine Cosine Algorithm (SCA), and Harris Hawk Optimization. The final feature subsets are combined to build a robust and compact input set. These same algorithms are then used to automatically optimize DL model architectures and hyperparameters. Experiments conducted on the RT-IoT 2022 dataset show that the SCA-optimized model achieved the best results, with 99.38% accuracy, 2.63% loss, and a consistent F1 score of 99.40% under stratified 5-fold cross-validation. These results are particularly notable given the imbalanced and complex nature of IoT traffic, and they indicate the model's ability to accurately detect both frequent and rare intrusion types. The proposed framework addresses the challenges of high-dimensional, heterogeneous IoT traffic by enabling fully automated, high-performance model design. These findings highlight its potential for real-world deployment in scalable and adaptive IoT intrusion detection systems. Future work will explore its applicability to broader IoT environments and integration with emerging paradigms such as federated and quantum machine learning.

1. Introduction

The Internet of Things (IoT) is a technology that enables devices to communicate with each other, making life easier, increasing efficiency, and creating smarter systems [1,2]. Nowadays, the integration of IoT technology into daily life allows these devices to be widely used [3,4]. In this context, it facilitates advancements in many smart areas, such as home systems, the healthcare sector, agriculture, industry and manufacturing, cities, and energy management [5–8]. IoT will have an even wider usage area in the future and will continue to offer innovative solutions in many sectors. The July 2024 report by IoT Analytics forecasts continued growth in spending on IoT technologies, including hardware, software, services, and security [9]. However, the growing number of cyberattacks targeting IoT devices highlights the need for more robust measures to protect such systems. These devices are often exposed to threats due to network security vulnerabilities, and those deployed in critical infrastructures pose particularly serious risks. According to SonicWall's mid-year 2024 threat report, IoT attacks have increased by a striking 107% year-on-year [10]. The company underlines that this increase is likely due to the fact that attackers are targeting devices more frequently, probably because IoT devices are easier targets. It is also emphasised that these devices often lack robust security measures. Due to the continuous increase in attacks, the development of more advanced intrusion detection systems (IDS) is becoming an important issue. For this purpose, it is of great importance to develop effective IDS for IoT devices using Artificial Intelligence (AI) methods. Methods such as machine learning (ML) and deep learning (DL), which are sub-disciplines of AI, are used in intrusion detection. ML enables computer systems to automatically perform and improve certain tasks by learning from data samples [11,12]. DL, a subfield of ML, leverages multilayer artificial neural networks to capture complex patterns and hierarchical representations within data [13]. These neural networks are organized in a way similar to the working principles of neurons in the

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human brain and are typically designed to work with large datasets. The main aim of this paper is to develop effective AI models for IDS.

IoT ecosystems introduce significant security and data privacy challenges due to their distributed, heterogeneous, and resource-constrained nature. The large number of interconnected devices increases the attack surface, making IoT systems highly vulnerable to cyber threats such as intrusion, data manipulation, and denial-of-service attacks. Furthermore, continuous data collection and transmission raise serious privacy concerns, particularly when sensitive user or system information is exposed. These challenges necessitate robust, scalable, and intelligent intrusion detection mechanisms capable of handling high-dimensional and dynamic IoT traffic patterns.

In this regard, the main contributions are summarized as follows:

(1) IoT systems generate high-dimensional, heterogeneous, and non-linear traffic patterns [14,15], making feature selection a critical step for effective intrusion detection. Feature selection improves deep learning performance by removing irrelevant and redundant attributes, reducing noise, mitigating overfitting, and lowering computational complexity, which is essential for real-time IoT environments. Unlike most prior studies that rely on either a single filter-based method or a single metaheuristic optimizer, this study introduces a synergetic two-stage hybrid feature selection framework. In the first stage, Random Forest (RF) is employed as a computationally efficient pre-filter to rank features and eliminate approximately 50% of low-importance attributes. This substantially reduces the search space and computational burden before entering the wrapper-based optimization stage. In the second stage, instead of depending on a single optimizer, five structurally diverse metaheuristic algorithms (GA, PSO, WOA, SCA, and HHO) are independently applied to refine the reduced feature set. The final subset is constructed using a consensus-based merging strategy that integrates overlapping and complementary features identified by different optimizers. This design mitigates optimizer-specific bias, enhances feature diversity, and improves robustness under imbalanced IoT traffic conditions.
(2) Beyond feature selection, this study extends metaheuristic optimization to deep learning architecture design. The refined feature set is used as input to DL models whose structural configurations and hyperparameters are automatically optimized using the same family of metaheuristic algorithms. This establishes a fully automated end-to-end pipeline that integrates feature engineering and model optimization, eliminating manual trial-and-error tuning commonly seen in prior IoT-IDS studies.
(3) Extensive experiments conducted on the publicly available RT-IoT 2022 dataset validate the effectiveness of the proposed framework under a 12-class classification setting. The SCA-optimized model achieved superior performance in terms of accuracy, F1-score, and loss, demonstrating strong generalization under stratified 5-fold cross-validation.
(4) The proposed framework provides a scalable, adaptable, and deployment-oriented solution for IoT intrusion detection. By unifying hybrid feature selection and automated DL optimization within a single structured architecture, it offers improved robustness, reduced dimensionality, and enhanced generalization in complex IoT environments.

The rest of the paper is structured as follows: Section 2 provides information on recent studies conducted on intrusion detection in IoT systems. Section 3 explains the dataset used, metaheuristic algorithms, and the proposed method. Section 4 presents the results observed from the simulations and the discussions regarding these findings. Finally, Section 5 concludes the study and outlines future research.

2. Related work

Recent studies have extensively investigated AI techniques for intrusion detection in IoT systems [17–23]. This section provides a literature review of these studies, highlighting their findings and positioning the contributions of this paper in comparison.

Saba et al. used a convolutional neural network (CNN) based model for IDS to detect abnormal behaviours in network traffic. By training this model on NID and BoT-IoT datasets, they obtained accuracy values of 99.51% and 92.85%, respectively. In their paper, while showing that vulnerabilities in IoT networks can be effectively detected, they also draw attention to challenges such as network complexity and resource constraints [24]. Kasongo developed IDS using different DL techniques such as Recurrent Neural Networks (RNNs), Long-Short Term Memory (LSTM) and Gated Recurrent Unit (GRU). He conducted tests on NSL-KDD and UNSW-NB15 datasets and obtained accuracy rates of 88.13% and 87.07%, respectively, thanks to FS with XGBoost algorithm [25]. For Medical IoT (MioT), one of the IoT application areas, Ioannou et al. proposed an energy efficient IDS. They present a security solution that provides low energy consumption and high accuracy in MIoT environments where devices operate with limited energy resources. In the three-stage system they proposed, in addition to detecting well-known attacks, they also detected unknown attacks using Federated Learning (FL). In the first stage, they achieved 99.98% accuracy rate with the Enhanced RF algorithm they chose as a classifier, while in the second stage, they achieved 99.7% accuracy rate with the One-Class Support Vector Machine (One-Class SVM) they used as an authenticator [26]. Turukmane et al. proposed a FS assisted IDS called Mud Ring assisted Multilayer Support Vector Machine (M-MultiSVM) for effective detection of network-based attacks. In their paper, they used CSE-CIC-IDS 2018 and UNSW-NB15 datasets. They solved the class imbalance in the datasets with the Advanced Synthetic Minority Oversampling Technique (ASmoT). They used Modified Singular Value Decomposition (M-SvD) for feature extraction to extract basic, content and traffic features from the dataset. The selected features were optimised with Opposition-based Northern Goshawk Optimisation (ONgO). In addition to the extracted and optimised features, hyperparameter tuning was performed with the Mud Ring optimization algorithm [27]. Nandanwar and Katarya proposed a model called AttackNet by combining CNN with GRU for Industrial IoT. The model was tested on the N_BaIoT dataset and achieved 99.75% accuracy, 99.75% precision and 99.74% recall, with an error rate of 0.0063 [28]. In another paper using the same dataset, they proposed TL-BILSTM IoT, a hybrid DL model, to detect botnet attacks in IoT environments [29]. Their proposed model combines CNN and Bidirectional Long Short-Term Memory (BiLSTM) structures with transfer learning techniques, aiming to detect BASHLITE and Mirai attacks across nine different IoT devices. According to experimental results, the model achieved 99.52% accuracy, 99.54% precision, 99.50% recall, and 99.23% F1-score, outperforming existing methods by 3.2% to 16.07%. Hossaion and Islam proposed a reliable IDS using an ML-based ensemble model. In this paper, the FS process was performed by correlation analysis, mutual information and Principal Component Analysis (PCA) methods. The model showed an accuracy rate of over 99%, especially with the RF algorithm, and high performance in other evaluation criteria [30]. Awajan presented DL-based IDS in his research. The proposed system works independently of the communication protocol and uses a fully connected Deep Neural Network (DNN) to detect anomalies in network traffic in real time. As a result of the tests, the average accuracy rate of the system was 93.74% [31]. Altulaihan et al. developed an IDS for detecting DoS attacks using the IoTID20 dataset. In their paper, they employed Correlation-Based Feature Selection (CFS) and GA for FS, alongside various ML methods. Decision Tree (DT) and RF, trained with features selected by GA, achieved the highest accuracy, with DT being found more efficient in terms of training and testing time [32]. Racherla et al. presented a DL-based IDS named Deep-IDS for real-time attack detection on IoT nodes. Using the CIC-IDS2017 dataset,

they trained an optimized LSTM network consisting of 64 hidden units. The system detects IoT attacks such as DoS, DDoS, Brute Force, Man-in-the-Middle, and Replay. A distinctive feature of their proposed method is the balance analysis between the Detection Rate (DR) and False Alarm Rate (FAR), which facilitates the real-time performance of Deep-IDS. Their method demonstrates a detection rate of 96.8% and an overall classification accuracy of 97.67% at a 70% threshold for the DR-FAR balance. Additionally, the Deep-IDS method exhibits superior performance with 97.67% precision, 98.17% recall, and a 97.91% F1-score [33]. Al-Ambusaidi et al. proposed ML-IDS, a system designed to ensure the security of IoT networks. They utilized the TON-IoT and UNSW-NB15 datasets to compare the performance of a modified RF algorithm with nine other ML algorithms for network attack detection. On the TON-IoT dataset, RF and other ensemble models (such as Gradient Boosting and Extreme Gradient Boosting) achieved 100% accuracy, precision, and specificity. On the UNSW-NB15 dataset, the RF algorithm demonstrated superior performance compared to many other methods, achieving an accuracy of 92.12% along with high precision and specificity values [34]. Hizal et al. suggested a two-stage DL-based IDS model for detecting DDoS attacks in IoT networks. They utilized the CICIoT2023 dataset to perform binary classification between attack traffic and normal traffic in the first stage, and then distinguished DDoS attacks from other attack types in the second stage. Additionally, they identified 10 subclasses of DDoS attacks. The performance of their proposed model was evaluated using various DL methods, including DNN, CNN, and LSTM. The best results were achieved with the LSTM-based two-stage model, which attained 94.96% accuracy and an F1-score of 92.32% [35]. Kaushik and Al-Raweshidy proposed a Teaching-Learning-Based Optimization (TLBO)-supported IDS. The system is designed to detect a wide range of attacks, including analysis, fuzzing, shellcode, worms, DoS, exploits, and backdoor intrusion attacks, while minimizing communication and device overhead. Using the UNSW-NB15 dataset, the model employs a RF classification algorithm optimized with TLBO. Experimental results revealed that the proposed TLBO-IDS outperforms state-of-the-art methods, such as the bat algorithm and GA. Compared to the bat algorithm, TLBO-IDS achieves a 7.1% higher detection rate and 11.4% better accuracy, while offering 40% lower communication overhead than GA, making it an effective and robust solution for IoT security [36]. Wang et al. presented a DL-based model for detecting multi-class network attacks in IoT networks. To address the issue of data imbalance, the model uses a Conditional Tabular Generative Adversarial Network (CTGAN) to generate synthetic data for minority classes. Additionally, it employs Group Convolution and Split-Attention structures to extract spatial features and BiGRU for capturing temporal features. The proposed method achieved impressive accuracy rates of 91.08%, 94.55%, and 97.47% on the UNSW-NB15, CIC-IDS2018, and CIC-IOT2023 datasets, respectively [37]. Hanafi et al. introduced a model based on Improved Binary Golden Jackal Optimization (IBGJO) and LSTM for detecting attacks in IoT networks. The proposed model was tested on the NSL-KDD and CIC-IDS2017 datasets. FS was performed using the IBGJO algorithm, while LSTM networks were employed to classify attack types. To address the local optima problem of the GJO algorithm and improve convergence speed, the IBGJO was enhanced using the Opposition-Based Learning (OBL) strategy. Results demonstrated that the IBGJO-LSTM model outperformed other methods, achieving an accuracy rate of 99.6%. The model effectively analyzed network traffic, delivering a higher detection rate and a lower false positive rate [38]. Sharma et al. developed a deep learning-based model for attack detection in IoT networks, incorporating Explainable AI (XAI) techniques to enhance the interpretability of the model. Their model was tested using the NSL-KDD and UNSW-NB15 datasets with DNN and CNN architectures. FS was performed using filter-based methods, achieving high accuracy with fewer features. The model demonstrated top performance with 99.4% accuracy on the NSL-KDD dataset (using 2D-CNN) and 81% accuracy on the UNSW-NB15 dataset. XAI methods, such as LIME and SHAP, were utilized to improve the interpretability of model predictions and to analyze which features contributed most significantly to the predictions [39]. An overview of the studies in the literature described and their comparative analyses are summarised in Table 1.

To the best of our knowledge, this study distinguishes itself from the existing literature by utilizing a two-stage FS framework, integrating RF-based filtering with multiple metaheuristic algorithms to achieve an optimized feature subset. Moreover, while hyperparameter optimization has been manually addressed in most prior studies, this paper employs metaheuristic algorithms to automate the optimization of the DL model's architecture, including the number of layers, activation functions, and learning rate, providing a systematic approach to IoT-related challenges.

3. Material and methods

This section covers the metaheuristic algorithms used for feature engineering and DL model optimization. In addition, the dataset used, the proposed method, and the performance evaluation criteria are discussed.

3.1. Metaheuristic algorithms

Metaheuristics play a significant role in FS as they can search for optimal solutions more effectively in large and complex datasets compared to traditional methods [40]. FS aims to eliminate redundant or irrelevant features to enhance a model's performance while reducing computational costs. These algorithms provide effective solutions for high-dimensional datasets, particularly due to their ability to explore global and local optima in the problem space. This subsection explains the metaheuristics used for FS.

3.1.1. Genetic algorithm (GA)

GA is a nature-inspired optimization method that seeks solutions by mimicking the fundamental mechanisms of biological evolution [41]. This method stands out for its ability to work effectively in complex and large solution spaces. It enables the evolution of individuals, representing potential solutions within a population, to produce better solutions across generations. The first step of the algorithm is to generate an initial population, either randomly or based on specific rules. Each individual is then evaluated using a predefined fitness function. Based on the fitness evaluation, the more successful individuals are selected to form the next generation. This selection process is typically performed using methods such as roulette wheel selection or tournament selection. Subsequently, the selected individuals combine their genetic information through a process called crossover to create new individuals. To increase diversity and prevent the algorithm from getting stuck in local optima, the genetic information of some individuals is randomly altered through a mutation process. The population composed of new individuals is used as the initial population for the next iteration. This process is repeated until a predefined stopping criterion is met. Leveraging the global search capability of GA, the most optimal feature combinations are discovered more efficiently. This enhances the model's accuracy while reducing computational costs by eliminating redundant features and mitigating the risk of overfitting.

3.1.2. Particle swarm optimization (PSO)

PSO is a metaheuristic algorithm developed by Kennedy and Eberhart, inspired by the social behavior of bird flocks [42]. Based on swarm intelligence, this method stands out for achieving faster convergence and being more cost-effective. In PSO, each solution is referred to as a particle within a swarm. Each particle has a specific velocity and position. These particles update their positions and velocities based on their own experiences and the experiences of their neighbors. The particle's previously best position is called the personal best, while the best position achieved by all particles in the swarm is known as the global best.

Table 1
Summary of current studies for IoT systems.

Year	Dataset	Feature selection	Hyperparameter tuning	Method	Paper
2022	BoT-IoT, NID	NaN	Manual	CNN	[24]
2023	NSL-KDD, UNSW-NB15	XGBoost-based	Manual	RNN, LSTM, GRU	[25]
2023	CSE–CIC–IDS2018, UNSW-NB15, NF-ToN-IoT, NSL-KDD, SIMARGL2021, WSN-DS, UKM-IDS20, NF-UQ-NIDS, Cyber Clean Center (CCC)	Correlation Analysis, Mutual Information, Principal Component Analysis	Manual	RF, Gradient Boosting, AdaBoost, XGBoost, Bagging, Stacking	[30]
2024	CSE-CIC-IDS 2018, UNSW-NB15	M-SvD, ONgO	Mud Ring Optimization Algorithm	M-MultiSVM	[27]
2024	N_BaIoT	NaN	Manual	AttackNet	[28]
2024	N_BaIoT	NaN	Manual	TL-BILSTM	[29]
2024	TON-IoT, UNSW-NB15	NaN	Manual	RF, Multi-Layer Perceptron, AdaBoost, Gradient Boosting, XGBoost, *k*-Nearest Neighbor (kNN), DT, Logistic Regression, Naive Bayes, Extra Tree	[34]
2024	CICIoT2023	Features with null or zero values were removed from the dataset and more meaningful features were used.	Manual	DNN, CNN, LSTM	[35]
2024	UNSW-NB15, CIC-IDS2018, CIC-IOT2023	Split-Attention	Manual	Residual Convolutional Neural Networks (ResNeSt), BiGRU	[37]
2024	NSL-KDD, CIC-IDS2017	IBGJO, BWOA, BSCA	Manual	LSTM	[38]
2024	NSL-KDD, UNSW-NB15	Correlation-Based Filter Method, Wrapper and Embedded methods	Manual	DNN, 1D-CNN, 2D-CNN	[39]

In this context, the velocity and position of each particle are continuously adjusted to align with the personal best and global best solutions, aiming to find an optimal solution. Consequently, the algorithm terminates when a predefined fitness value is achieved. From a mathematical perspective, in a D-dimensional space, the positions and velocities of N particles are initialized randomly, as in all metaheuristic algorithms. Eq. (1) presents the solution of the ith particle at iteration t. According to Eqs. (2) and (3), the current solution (for each particle) is updated based on the local and global optima.

$$Q_i^t = \left[q_{i,1}^t, q_{1,j}^t, \ldots, q_{i,d}^t\right] \; i-1,2,\ldots.N \tag{1}$$

$$V_{i,j}^{t+1} = wv_{i,j}^t + C_1 r_1 \left(\text{Pbest}_{i,j}^{\,t} - q_{i,j}^t \right) + C_2 r_2 \left(\text{Gbest}_{\,j}^{\,t} - q_{i,j}^t \right) \tag{2}$$

$$q_{i,j}^{t+1} = q_{i,j}^t + V_{i,j}^{t+1} \tag{3}$$

where, C_1 and C_2 are learning factors used in the velocity updates of the particles. These factors determine the attraction strength of a particle toward its personal best position and the global best position. r_1 and r_2 are variables that take random values in the range [0,1]. w, referred to as the inertia weight, represents the extent to which a particle retains its current velocity (Eq. (4)).

$$w = w_{max} - \text{ iteration} \cdot \left(\frac{w_{max} - w_{min}}{\max_{\text{ -iteration}}} \right) \tag{4}$$

3.1.3. Whale optimization algorithm (WOA)

WOA is a stochastic, population-based algorithm developed by mimicking the bubble-net feeding strategy exhibited by humpback whales during hunting [43]. This metaheuristic consists of two phases: exploitation, which involves encircling the prey and the spiral bubble-net attack method, and exploration, representing the random search process for prey. The explanations for these phases are provided below:

3.1.3.1. Exploitation phase. To update the movement of a solution, the mathematical representation of the whale's encircling behavior, as described by Eq. (5) and Eq. (6), is used.

$$\overrightarrow{D} = \left| \overrightarrow{C} \cdot \overrightarrow{X^*}(t) - \overrightarrow{X}(t) \right| \tag{5}$$

$$\overrightarrow{X}(t+1) = \overrightarrow{X^*}(t) - \overrightarrow{A} \cdot \overrightarrow{D} \tag{6}$$

where, t represents the current iteration, and X^* is the best solution obtained up to iteration t. While X denotes the current solution, A and D are coefficient vectors, which are calculated using Eqs. (7) and (8), respectively.

$$\overrightarrow{A} = 2\overrightarrow{a} \cdot \overrightarrow{r} - \overrightarrow{a} \tag{7}$$

$$\overrightarrow{C} = 2 \cdot \overrightarrow{r} \tag{8}$$

where, r is a vector that takes random values in the range [0,1]. The parameter a decreases linearly from 2 to 0, reflecting the humpback whales' behavior of narrowing their encircling mechanism while moving towards their prey and following a spiral-shaped path. This path is determined by calculating the distance between the solution (X) and the best solution (X^*) (Eqs. (9) and 10). Adjustments to the position based on the known best solution are updated in Eq. (6). A and C control the regions where a solution might be located around the best solution.

$$\overrightarrow{X}(t+1) = \overrightarrow{D'} \cdot e^{bl} \cdot \cos(2\pi l) + \overrightarrow{X^*}(t) \tag{9}$$

$$\overrightarrow{D'} = \left| \overrightarrow{X^*}(t) - \overrightarrow{X}(t) \right| \tag{10}$$

where, l is a stochastic variable that takes values in the range $[-1, 1]$. The parameter b is a constant that defines the logarithmic spiral structure. Element-wise multiplication is represented by the $\cdot$ symbol. D' denotes the distance between the ith whale and the prey, which is the optimal solution obtained up to the current iteration. To mathematically represent the shrinking encircling mechanism and the upward spiral path, a 50% probability is used to alternate between these behaviors during optimization. This is expressed by Eq. (11), where p is a random number between 0 and 1.

$$\overrightarrow{X}(t+1) = \begin{cases} \text{ShrinkingEncircling } (Eq.6) \text{ if } (p < 0.5) \\ \text{spiral } - \textit{shapedpath } (Eq.9) \text{ if } (p \geq 0.5) \end{cases} \tag{11}$$

3.1.3.2. Exploration phase. In WOA, position updates are performed using a randomly selected solution. If $\overrightarrow{A}$ is greater than 1 or less than -1, it is used to include random values. Subsequently, a solution is made to

move away from the best-known search agent. The mathematical modeling of this behavior is given in Eqs. (12) and (13). $\vec{X}_{rand}$ represents a randomly selected whale in the current iteration.

$$\vec{D} = \left|\vec{C}\cdot\vec{X}_{rand} - \vec{X}\right|\cdot\vec{D} \tag{12}$$

$$\vec{X}(t+1) = \vec{X}_{rand} - \vec{A} \tag{13}$$

3.1.4. Sine cosine algorithm (SCA)

SCA is a stochastic metaheuristic inspired by mathematically based sine and cosine functions [44]. It begins by working with a set of solutions or search agents randomly positioned in the search space of an optimization problem. In each iteration, search agents are guided toward an optimal solution within the search space using a fitness function that evaluates the suitability of each agent. Here, each agent is defined by an n-dimensional decision variable vector. This metaheuristic employs a population consisting of m search agents, where each agent is defined by an n-dimensional decision variable vector. The vector is represented as $X_i = (x_{i1}, x_{i2}, \ldots, x_{in})$, where X_i denotes the ith search agent in the population. Additionally, the algorithm tracks the position of the best solution (P) in each iteration. The mathematical expression for the update function used for X_i is presented in Eqs. (14) and (15).

$$x_{ij}^{t+1} = \begin{cases} x_{ij}^{t} + r_1 * sin\,(r_2) * \left|r_3 P_j^t - x_{ij}^t\right| & \text{if } s < 0.5 \\ x_{ij}^{t} + r_1 * cos\,(r_2) * \left|r_3 P_j^t - x_{ij}^t\right| & \text{if } s \geq 0.5 \end{cases} \tag{14}$$

$$r_1 = a - t\frac{a}{T_{max}} \tag{15}$$

where, t represents the current iteration. r_1 is a variable that linearly decreases from a constant a to 0, ensuring the balance between exploration and exploitation. The other symbols in the equations represent random numbers.

This mathematical expression applies a circular search pattern, where the best solution or target point is located at the center of a circle, and the search agents are positioned around this center. The circular search area is divided into subregions representing possible exploration zones for X_i. The r_1 determines the movement of X_i toward these regions. If $r_1 > 1$, moves toward P, representing the exploitation phase. If $r_1 \leq 1$, X_i moves away from P, transitioning to the exploration phase. This parameter decreases linearly, establishing a balance between the exploration and exploitation processes (Eq. (15)). The random r_2 value determines how far X_i will move in the direction defined by r_1. The r_3 assigns random weights to the best solution P; when $r_3 > 1$, the influence of P is increased, while it is reduced when $r_3 < 1$. Additionally, the s enables random switching between sine and cosine functions.

3.1.5. Harris hawks optimization (HHO)

HHO is a more recent algorithm compared to the metaheuristics used. It was developed by Heidari et al., inspired by the hunting behaviors of Harris hawks [45]. Like all metaheuristics, this algorithm begins by generating a randomly initialized solution space. It consists of two main phases: exploration and exploitation. The exploration phase mimics the behaviors of hawks searching for prey, i.e., solutions. Typically, two strategies are employed to explore the search space: discovering a new search area based on the positions of other hawks and conducting random exploration within the current range. The mathematical model of this exploration process, which assumes equal selection probabilities, is presented in Eq. (16).

$$X(t+1) = \begin{cases} X_{\text{rand}}\,(t) - r_1\left|X_{\text{rand}}\,(t) - 2r_2 X(t)\right| & q \geq 0.5 \\ \left(X_{\text{rabbit}}\,(t) - X_m(t)\right) - r_3\left(LB + r_4(UB - LB)\right) & q < 0.5 \end{cases} \tag{16}$$

where, X is the position vector, and $X(t+1)$ represents the next position of the hawks for the t-th iteration. $X_{\text{rabbit}}\,(t)$ and $X_{\text{rand}}\,(t)$ denote the position of the prey and a randomly selected hawk at iteration t, respectively. X_m is the average position of all hawks. The lower and upper bounds are represented by LB and UB, respectively. The remaining variables take random values in the range [0, 1]. To calculate the average positions of the hawks at the tth iteration, Eq. (17) is used, where N represents the number of hawks.

$$X_m(t) = \frac{1}{N}\sum_{i=1}^{N} X_i(t) \tag{17}$$

The mathematical model that calculates the escape energy of prey at a given iteration is presented in Eq. (18). Here, E_0 is the initial energy of the prey, which takes random values in the range $[-1, 1]$. If the absolute value of E is greater than or equal to 1, the algorithm transitions to the exploration phase; if it is <1, the exploitation phase is initiated.

$$E = 2\ \times E_0\left(1 - \frac{t}{T}\right) \tag{18}$$

The HHO employs two methods, soft besiege and hard besiege, to perform the exploitation phase linked to E. The transition between these methods is based on a random variable r, which simulates the probability of the prey escaping from the hawk. The likelihood of a successful escape is controlled by a threshold defined by the condition $r \geq 0.5$. When $r \geq 0.5$ and $|E| \geq 0.5$, soft besiege occurs. This indicates that the prey has high energy and is mathematically expressed as shown in Eq. (19). $\Delta X(t)$ represents the difference between the prey's current position and the position vector at iteration t. Additionally, $J = 2(1-r_5)$ denotes a random escape leap, calculated using r_5, a random value in the range [0, 1].

$$X(t+1) = \Delta X(t) - E\left|JX_{\text{rabbit}}\,(t) - X(t)\right| \tag{19}$$

$$\Delta X(t) = X_{\text{rabbit}}\,(t) - X(t) \tag{20}$$

The hard besiege phase occurs when the prey's energy level is below the threshold value and is mathematically presented in Eq. (21). During this phase, hawks reduce the distance between themselves and the prey, i.e., the solutions. Y and Z are determined according to Eqs. (22) and 23. Here, S represents a random vector, while LF denotes the Levy flight function.

$$X(t+1) = \begin{cases} Y \text{ if } F(Y) < F(X(t)) \\ Z \text{ if } F(Z) < F(X(t)) \end{cases} \tag{21}$$

$$Y = X_{rabbit}(t) - E\left|JX_{rabbit}(t) - X_m(t)\right| \tag{22}$$

$$Z = Y + S \times LF(D) \tag{23}$$

3.2. Solution representation for feature selection

Metaheuristic algorithms provide a powerful approach for selecting optimal feature subsets in high-dimensional datasets. These algorithms aim to eliminate irrelevant or redundant features and optimize model performance through their global search capabilities. Solutions are typically represented by a binary vector, where each feature is assigned a value of "1″ if selected or "0" if not (**see** Fig. 1). This representation

f_1	f_2	f_3	f_4	f_5	...	f_n
0	1	1	0	1	...	1

☐ Selected feature
☐ Not selected feature

Fig. 1. Solution illustration as a binary vector.

allows the algorithms to effectively explore the solution space and identify meaningful features to enhance model performance.

3.3. Evaluation metrics

Evaluation metrics are crucial both during the FS process and in comparing the performance of ML models. These metrics are calculated using the confusion matrix. Table 2 provides details and explanations of these metrics.

3.3.1. Fitness function

The fitness function plays a critical role in the FS process by ensuring the identification of the most optimal features, while enhancing both the accuracy and efficiency of ML algorithms. The objective in FS is to maximize classification accuracy as close to 100% as possible while minimizing the number of selected features. Accordingly, the mathematical equation for the objective function is presented in Eqs. (24) and (25).

$$Error_{rate} = 1 - Accuracy \tag{24}$$

$$f_{fitness} = \alpha * \ Error_{rate} + \beta * \frac{F_s}{F_a} \tag{25}$$

where, $Error_{rate}$ is determined based on the classification accuracy of the kNN algorithm (where $k = 5$). kNN is a popular ML algorithm that belongs to the supervised learning category and is used for both classification and regression problems [46]. F_a and F_s represent the total number of features in the dataset and the number of selected features, respectively. α and β are weight coefficients in the fitness function that determine the importance of the two main objectives: classification accuracy and the number of selected features, with their values set to 0.99 and 0.01, respectively.

3.4. Binary mechanism

Metaheuristics convert continuous solution space values into binary format using a threshold-based mechanism (Eq. (26)). This approach facilitates the elimination of irrelevant features and the identification of relevant ones by comparing continuous values against a threshold. In this paper, chromosome genes in GA, particle velocities in PSO, continuous positions in WOA and SCA, and hawk positions in HHO were transformed into binary space based on a predefined threshold. This method represents an effective mechanism for converting continuous solution values into binary values.

$$x_{\text{binary}} = \begin{cases} 1, & \text{if } x \geq \eta \\ 0, & \text{if } x < \eta \end{cases} \tag{26}$$

where, the transformed binary solution obtained through FS is referred to as x_{binary} . If this solution is 1, it indicates that the feature is selected; otherwise, it is not selected (see Fig. 1). x represents a value in the continuous solution space, and η is the threshold value, set to 0.5.

3.5. RT-IoT 2022 dataset

The RT-IoT 2022 dataset, presented by Sharmila and Nagapadma, is a specialized dataset obtained from a real-time IoT infrastructure, encompassing both normal and malicious network behaviors to broadly reflect real-world scenarios [16]. The dataset integrates data collected from IoT devices such as ThingSpeak-LED, Wipro-Bulb, and MQTT-Temp, as well as simulated attack scenarios including Brute-Force SSH attacks, Hping and Slowloris-based DDoS attacks, and Nmap patterns. This provides a detailed representation of the complex nature of network traffic. The dataset consists of 12 classes, and their distribution is illustrated in Fig. 2.

3.6. Proposed method

In this paper, a novel two-stage FS and optimization framework are proposed to enhance the performance of DL models by selecting the most relevant features and optimizing model configurations effectively. The proposed feature selection process follows a two-stage hybrid strategy combining filter-based preselection and wrapper-based optimization. The method comprises the following sequential steps:

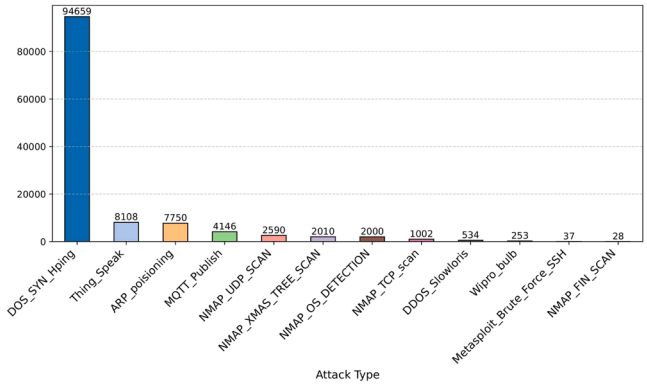
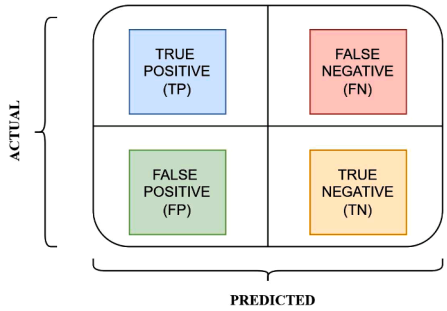


Fig. 2. Class distribution of the dataset.

Table 2
Evaluation metrics and details.

Confusion matrix	Metric	Equation	Explanation
	Accuracy	$\frac{TP + TN}{TP + TN + FP + FN}$	Measures the proportion of correctly classified instances, assessing the overall performance of the model.
	Recall	$\frac{TP}{TP + FN}$	Evaluates the proportion of actual positives correctly identified by the model.
	Precision	$\frac{TP}{TP + FP}$	Measures the proportion of predicted positives that are actually positive.
	F_1 score	$\frac{2 \times (\text{Precision} \times \text{Recall})}{(\text{Precision} + \text{Recall})}$	Represents the harmonic mean of precision and recall, balancing their trade-off for overall performance evaluation.

TP: The model correctly predicts positive instances that are actually positive, **FP:** The model incorrectly predicts instances as positive when they are actually negative (false alarm), **FN:** The model incorrectly predicts instances as negative when they are actually positive (missed positives), **TN:** The model correctly predicts negative instances that are actually negative.

- The initial stage of the two-stage framework leverages RF to perform an efficient preliminary feature selection. This crucial step reduces the dataset's dimensionality by roughly 50% by eliminating the least relevant features, which dramatically shrinks the search space for the metaheuristic algorithms in the second stage. This strategic reduction not only saves significant computational time but also helps the metaheuristics find a more optimal solution. This reduction also improves deep learning performance by minimizing noise, reducing overfitting, and enabling more efficient model training. RF was chosen as the first-stage filter method due to its ability to efficiently rank features by importance, robustness to noise and imbalance, and interpretability in the IoT security context. Unlike linear filters (e.g., correlation) or dimensionality reduction methods (e.g., PCA), it maintains feature semantics and captures non-linear interactions, which are critical for intrusion detection. Furthermore, it requires minimal parameter tuning and scales effectively to high-dimensional traffic data, making it a practical pre-filter before applying metaheuristic optimization.
- In the second stage, five metaheuristic algorithms are independently applied to the reduced dataset obtained from the RF process. These algorithms each leverage their unique search strategies to identify optimal feature subsets: GA utilizes a robust global exploration via crossover and mutation, while PSO achieves rapid convergence through social information sharing, WOA employs an adaptive search, SCA maintains an efficient balance between exploration and exploitation, and HHO applies dynamic besiege strategies. Instead of relying on a single metaheuristic, the proposed framework merges the feature subsets generated by these different algorithms. This approach leverages their complementary search strategies to increase the likelihood of capturing diverse and relevant features. By integrating both overlapping and unique features from the multiple optimizers, the resulting subset becomes both compact and comprehensive, thereby improving the robustness and generalization of the intrusion detection model.
- The proposed DL models were automatically optimized within a well-defined, performance-sensitive hyperparameter space by the same five metaheuristic algorithms used in the FS stage. Each candidate architecture was encoded as an "individual," whose genes specified structural and training hyperparameters such as layer types and counts, units or filters per layer, kernel sizes, activation functions, optimizers, learning rates, and Dropout/BatchNorm settings. During each iteration, candidate models were trained and evaluated on the test set, with test-set accuracy serving as the fitness function to select the highest-performing individuals for the next generation. This fully automated search discovered optimal configurations, enabling flexible adaptation to IoT security tasks such as anomaly detection and threat prediction. As a result, the parameter selection process was fully automated. It was directly guided by model performance, producing configurations that were both diverse and generalizable. Flow diagram of the proposed method is presented in Fig. 3.

During the data preprocessing stage, categorical features were converted into numerical representations, and the dataset was normalized using the MinMax Scaler to ensure all features are scaled to a uniform range, facilitating better performance in subsequent modelling steps. In all stages of the process, the dataset was split into 70% for training and 30% for testing. Following the RF feature importance assessment, the features with the highest importance scores were identified and are shown in Fig. 4. As shown in Fig. 4, the features obtained were further refined through metaheuristic-based FS running for 100 iterations. To enhance both FS and model optimization, the proposed framework employs a diverse set of five metaheuristic algorithms. Each metaheuristic algorithm exhibits distinct search dynamics, including rapid convergence (PSO), strong global exploration (GA), adaptive encircling mechanisms (WOA), oscillatory exploitation–exploration balance (SCA), and dynamic besiege strategies (HHO). These differences allow the algorithms to traverse different regions of the reduced feature space obtained after the RF prefiltering stage. The merging process is performed using a union-based aggregation strategy applied to the independently selected feature subsets. Since all metaheuristics operate on the already reduced RF-filtered space, the final merged subset remains significantly smaller than the original feature set (overall reduction >70%). Moreover, each optimizer is guided by a fitness function that jointly maximizes classification accuracy while penalizing excessive feature counts, thereby encouraging compact subsets before merging. The union strategy preserves both consensus

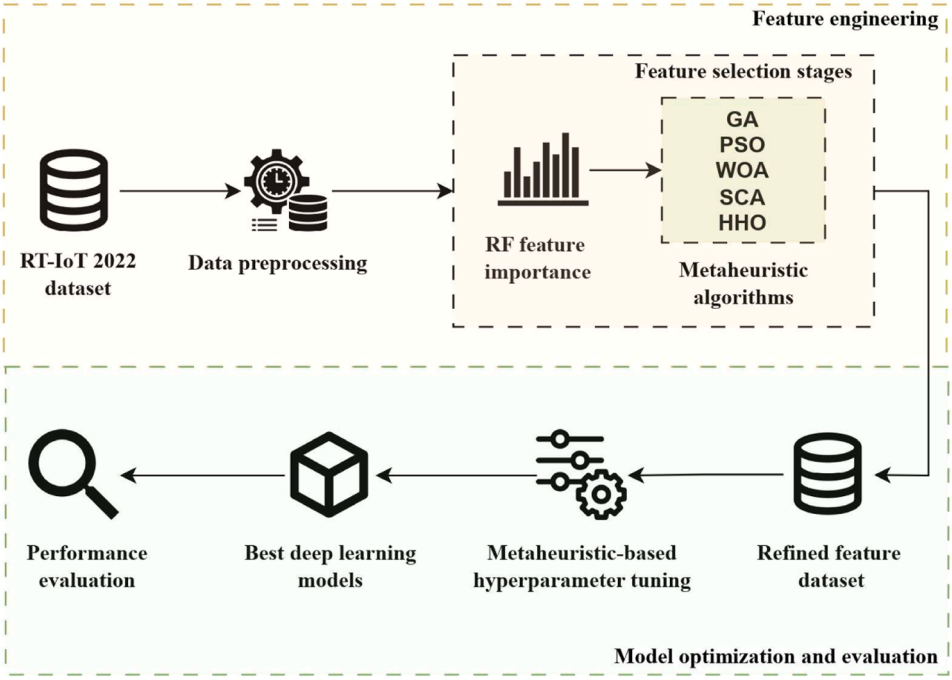


Fig. 3. Flow diagram of the proposed method.

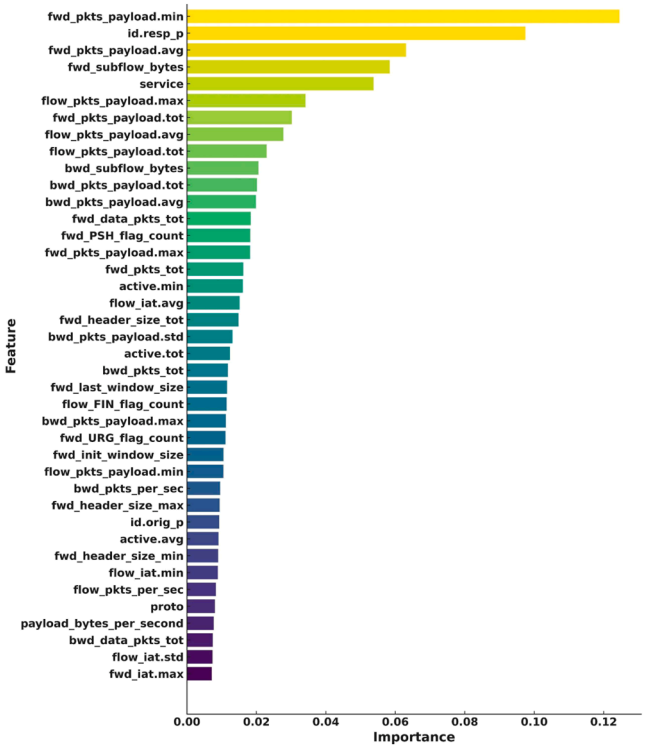


Fig. 4. Feature importance scores.

features (selected by multiple optimizers) and complementary features (identified by specific search trajectories), preventing optimizer-specific bias while maintaining diversity. Importantly, redundancy is implicitly controlled through the two-stage selection pipeline: RF removes low-importance features in the first stage, and wrapper-based optimization further eliminates irrelevant attributes based on classification performance. Table 3 presents the selected features and the algorithms that selected them. These features were used to develop DL models. In addition, the convergence plots of the fitness values are presented in Fig. 5.

After the completion of the two-stage FS process, a comprehensive and well-structured hyperparameter search space was defined to guide the optimization of DL models. As summarized in Table 4, this search space encompasses a diverse set of architectural and training-related parameters, allowing metaheuristic algorithms to explore a wide range of candidate configurations. The core architecture of the models is modular and flexible, beginning optionally with an LSTM layer to

Table 3
Features selected by each metaheuristic algorithm.

Feature	Selecting algorithm	Feature	Selecting algorithm
fwd_pkts_payload.min	WOA, HHO	id.resp_p	GA, SCA
service	PSO, HHO	flow_pkts_payload.avg	PSO
fwd_data_pkts_tot	WOA	fwd_PSH_flag_count	HHO
fwd_pkts_payload.max	GA, PSO, SCA	active.min	HHO
flow_iat.avg	WOA	bwd_pkts_payload.std	PSO, WOA
bwd_pkts_tot	SCA	fwd_last_window_size	SCA
flow_FIN_flag_count	GA, HHO	bwd_pkts_payload.max	WOA, HHO
fwd_URG_flag_count	WOA, SCA	flow_pkts_payload.min	WOA
fwd_header_size_max	GA, PSO, HHO, SCA	id.orig_p	PSO, WOA
active.avg	SCA	fwd_header_size_min	PSO, WOA
flow_iat.min	GA	*flow_pkts_per_sec*	PSO

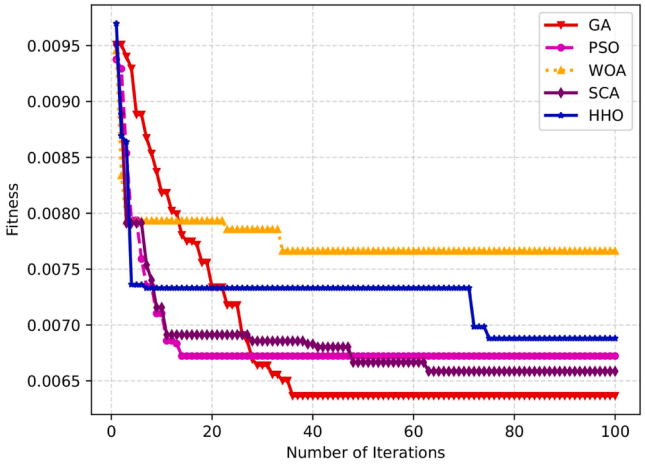


Fig. 5. Convergence plots of fitness values for metaheuristic algorithms.

capture temporal dependencies, followed by a CNN layer for spatial and hierarchical feature extraction, and concluding with a fully connected ANN layer to consolidate learned representations for final prediction. Each of these components can be selectively included or excluded, based on optimization decisions, enabling dynamic model composition tailored to the data characteristics. To further enhance performance and generalization, optional Batch Normalization (excluded for ANN layers) and Dropout are applied after each layer. If selected, Dropout is implemented with fixed probabilities of 0.25 per layer and 0.30 for general dropout. These binary options ([0, 1]) allow metaheuristic algorithms to activate or deactivate regularization mechanisms as part of the optimization process. The hyperparameter ranges were not arbitrarily chosen; rather, they were grounded in well-established practices from the DL and intrusion detection literature, and refined through empirical validation. For instance, the unit ranges for LSTM and ANN layers ([32, 256]) and the filter ranges for CNNs ([32, 128]) offer a balance between lightweight and expressive architectures, ensuring compatibility with the resource constraints typical of IoT environments. Similarly, the learning rate range ([0.0001, 0.01]) reflects a standard interval known to support stable convergence across diverse training conditions. The inclusion of multiple activation functions and optimizers (e.g., ReLU, ELU, Adam, Nadam, etc.) further enables the discovery of high-performing configurations without manual intervention. By defining this rich and performance-sensitive parameter space, it ensures that the metaheuristic search process is both efficient and comprehensive; it can adapt the architectural depth, layer composition, training dynamics, and tuning strategies to specific characteristics of the dataset.

4. Results and discussion

This section presents the experimental results of the proposed DL models and compares their performance with findings reported in the literature. The aim is to evaluate the effectiveness of the developed approach in relation to previous studies. These findings, together with our practical considerations and limitations, confirm the approach's applicability and outline its constraints.

4.1. Experimental results

The results evaluated in the test dataset using various performance metrics are summarised in Table 5, which demonstrates SCA's superior performance across all evaluated metrics for DL model optimization. Achieving the lowest loss value, SCA outperformed WOA by 0.0023 (approximately 8% less error), and showed even greater improvements compared to PSO (0.0089), GA (0.0315), and HHO (0.0171).

SCA also achieved the highest accuracy, exceeding WOA by 0.15%,

Table 4
Parameter space and optimal values determined by metaheuristic algorithms.

Parameter	Range/Space	Optimum Value				
		GA	PSO	WOA	SCA	HHO
LSTM layer	[0, 2]	1	0	1	0	1
CNN layer	[1, 3]	3	2	1	2	2
ANN layer	[1, 3]	1	3	1	2	2
LSTM unit	[32, 256]	68	NaN	213	NaN	32
CNN filter	[32, 128]	Layers 1, 2, 3 = 68, 99, 81	Layers 1, 2 = 50, 64	56	Layers 1, 2 = 35, 103	Layers 1, 2 = 77, 80
CNN kernel	[2, 5]	Layers 1, 2, 3 = 4, 5, 2	Layers 1, 2 = 3, 2	3	Layers 1, 2 = 3, 3	Layers 1, 2 = 3, 2
ANN unit	[32, 256]	185	Layers 1, 2, 3 = 136, 209, 239	71	Layers 1, 2 = 184, 98	Layers 1, 2 = 133, 32
Activation function for CNN	"relu", "leaky_relu", "selu", "elu", "softplus", "tanh"	Layers 1, 2, 3 = "leaky_relu", "softplus", "softplus"	Layers 1, 2 = "relu", "selu"	"relu"	Layers 1, 2 = "elu", "leaky_relu"	Layers 1, 2 = "elu", "elu"
Activation function for ANN		"leaky_relu"	Layers 1, 2, 3 = "softplus", "elu", "elu"	"elu"	Layers 1, 2 = "tanh", "leaky_relu"	Layers 1, 2 = "selu", "leaky_relu"
Optimizer	"adam", "SGD", "RMSprop", "Adagrad", "Adadelta", "Nadam"	"Nadam"	"Nadam"	"Nadam"	"adam"	"RMSprop"
Learning rate	[0.0001, 0.01]	0.001,611	0.001,531	0.003,096	0.00,025	0.00,526
Batch normalization for LSTM	[0, 1]	0	NaN	0	NaN	1
Batch normalization for CNN		Layers 1, 2, 3 = 1, 1, 1	Layers 1, 2 = 1, 1	0	0	Layers 1, 2 = 0, 1
Dropout for LSTM	[0, 1]	0	NaN	0	NaN	0
Dropout for CNN		Layers 1, 2, 3 = 0, 1, 1	0	0	0	Layers 1, 2 = 0, 1
Dropout for ANN		0	Layers 1, 2, 3 = 1, 1, 1	0	0	Layers 1, 2 = 0, 1
General dropout	[0, 1]	0	1	0	0	0

Table 5
Results obtained according to various performance criteria.

Metric	GA	PSO	WOA	SCA	HHO
Loss	0.0578	0.0352	0.0286	**0.0263**	0.0434
Accuracy	0.9856	0.9909	0.9923	**0.9938**	0.9854
Recall	0.9855	0.9909	0.9921	**0.9937**	0.9853
Precision	0.9861	0.9910	0.9924	**0.9939**	0.9856
F1 Score	0.9858	0.9910	0.9923	**0.9938**	0.9855

PSO by 0.29%, GA by 0.82%, and HHO by 0.84%. Consistent with these results, SCA led in recall (0.16% higher than WOA, 0.28% higher than PSO, and approximately 0.8% higher than both GA and HHO) and precision (0.15% over WOA, 0.29% over PSO, 0.78% over GA, and 0.83% over HHO). This consistent trend continued with the F1 Score, where SCA held a 0.15% advantage over WOA, a 0.28% advantage over PSO, and approximately 0.8% advantages over both GA and HHO. These results confirm SCA's consistent outperformance across all metrics, making it the most effective and reliable algorithm for DL model optimization in this paper. The training and validation convergence plots provide insight into the performance of all models across epochs. Fig. 6 presents the accuracy and loss curves for both training and validation phases. Among the evaluated models, the one optimized by SCA shows better convergence and stability compared to the others. The superior performance of SCA can be attributed to its dynamic sine–cosine position updating strategy, which effectively balances global exploration and local exploitation. Consequently, this mechanism enables smoother convergence and reduces the risk of being trapped in local minima during deep learning hyperparameter optimization.

In the proposed framework, SCA operates on a reduced and more relevant feature space obtained after the RF-based pre-selection stage, allowing it to efficiently focus on high-quality regions of the search space. This results in improved model generalization and performance. These findings are reflected in the achieved highest accuracy and lowest loss values among all tested algorithms. Fig. 7 shows the optimized models' confusion matrices.

These matrices offer a detailed breakdown of true positives, false positives, true negatives, and false negatives, providing a comprehensive evaluation of each algorithm's predictive effectiveness. The confusion matrices reveal distinct classification patterns across the different algorithms, with some classes being handled remarkably well and others presenting challenges. "DOS_SYN_Hping", "NMAP_FIN_SCAN", and "NMAP_OS_DETECTION" were perfectly classified by all models, demonstrating the algorithms' strong generalization capabilities for these well-represented classes. Similarly, "MQTT_Publish" and "NMAP_XMAS_TREE_SCAN" showed only minor misclassifications, with all algorithms performing nearly equally well. In contrast, other classes revealed notable variations in performance. For "ARP_Poisoning", models optimized by PSO, SCA, and HHO outperformed those by GA and WOA, achieving better classification accuracy. In the case of "DDOS_Slowloris", GA demonstrated the best classification performance, outperforming all other algorithms. Meanwhile, "Metasploit_Brute_Force_SSH" proved highly challenging for all algorithms, with PSO and HHO failing entirely to classify this class correctly, and other algorithms performing only moderately well, likely due to the limited number of samples for this category. For "NMAP_TCP_scan", all algorithms achieved relatively similar classification accuracy, showcasing their consistent ability to handle this class effectively. In "NMAP_UDP_SCAN", PSO and SCA delivered the best performance, minimizing misclassifications compared to the other algorithms. For "Thing_Speak", WOA emerged as the top performer with the least misclassifications, while HHO struggled significantly, misclassifying the most instances. Lastly, in "Wipro_bulb", HHO outperformed all other algorithms, achieving the highest classification accuracy.

Stratified K-Fold (SKFold) cross-validation was implemented to ensure a balanced evaluation of the models across all classes, particularly given the significant class imbalance in the dataset [47]. SKFold divides the data into stratified folds, maintaining the class distribution in each fold to reflect the original dataset. In our paper, stratified 5-fold cross-validation was applied to achieve this balance. This approach is commonly used to prevent biases caused by underrepresentation or overrepresentation of certain classes, enabling a more reliable assessment of the models' performance. Additionally, SKFold helps to evaluate the models' generalization capabilities and identify potential overfitting

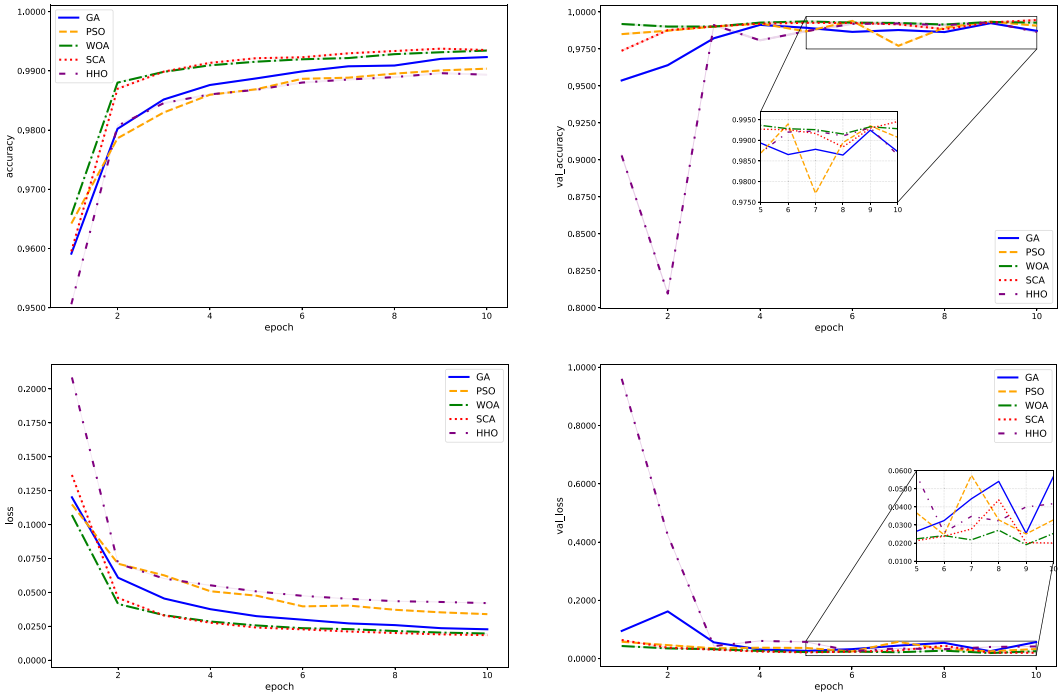


Fig. 6. Training and validation convergence plots for metaheuristic-optimized DL models.

by testing them on multiple, diverse subsets of the data. The average values obtained after 5-fold cross-validation are summarized in Table 6, providing a comprehensive evaluation of the models' performance. When Table 6 is analysed, the model generated by SCA gave better results in terms of generalisation ability. On the contrary, PSO's model was less successful compared to the others. WOA was one of the best performing algorithms following SCA. GA and HHO proved to produce general models with similar good results. Furthermore, Fig. 7 presents the confusion matrices of the trained models, providing a detailed class-wise evaluation and confirming the stability of predictive behavior across both majority and minority attack categories.

4.2. Comparison with existing literature

This subsection compares the performance of the proposed models with results reported in previous studies. Table 7 summarises the comparison between the proposed method and existing studies.

According to Table 7, the proposed Two-stage FS + SCA-DL framework emerges as a highly competitive alternative to the approaches found in the existing IoT-IDS literature. While many earlier studies rely on well-known public datasets and focus on 2- to 15-class tasks, our proposed model is evaluated on the complex 12-class RT-IoT 2022 dataset and achieves a superior accuracy of 99.38%. Specifically, when compared to the work of Hizal et al., which also addressed a 12-class classification task using a two-stage LSTM, our framework demonstrates a significant performance improvement of over 8% (99.38%vs. 91.27%). Furthermore, the robustness of our metaheuristic-driven approach is evident when contrasted with high-class count models such as Wang et al., who reported accuracies of 94.55% and 97.47% for 15-class and 13-class tasks, respectively. Even against traditional feature selection methods, such as the XGBoost-based framework by Kasongo, which saw accuracy drop to 78.40% for 10-class scenarios, our model maintains exceptional stability. These results, corroborated by high F1 scores and low loss values under stratified 5-fold validation, confirm that the proposed framework handles high-dimensional and imbalanced IoT traffic more effectively than existing state-of-the-art methods. Firstly, this superior performance derives from our two-stage feature selection strategy. In the initial filtering stage, a RF ranks and eliminates noisy or redundant IoT traffic attributes, sharply reducing dimensionality while preserving the most informative features. In the subsequent wrapper stage, five metaheuristic optimizers refine this compact set: GA drives broad global exploration through crossover and mutation; PSO accelerates convergence via cooperative velocity updates; WOA applies adaptive bubble-net encircling for focused local exploitation; HHO intensifies global search using dynamic prey–predator behaviors; and the SCA employs oscillatory sine and cosine movements to efficiently escape local optima. The complementary strengths of these methods form a highly distinctive set of features that support the model's exceptional accuracy. Second, the optimized feature subset serves as input to our DL classifier, with hyperparameters such as learning rate, network depth, and regularization strength automatically tuned by the same five metaheuristic optimizers. Each algorithm configures the model end-to-end without manual intervention, ensuring unbiased optimization across folds. Notably, the SCA outperforms the others. These results stem from SCA's simple sine–cosine update rules, which require only four control parameters and feature adaptive oscillations that balance exploration and exploitation to avoid local optima [44,50]. The oscillatory mechanism increases search diversity and enables efficient navigation of the high-dimensional, non-convex loss landscape of DL models. In addition, SCA's lightweight parameterization reduces tuning complexity and improves stability. These strategies may explain why SCA achieved faster convergence and superior accuracy in our experiments.

4.3. Practical implications and limitations

The empirical results, including a 99.38% accuracy and consistently low cross-validated loss, indicate that the proposed framework effectively discriminates between benign and malicious IoT traffic across multiple attack categories. The high true-positive rates and controlled false-positive levels suggest stable detection performance under imbalanced multi-class conditions. Unlike fixed-architecture or single-optimizer approaches, the proposed framework integrates hybrid

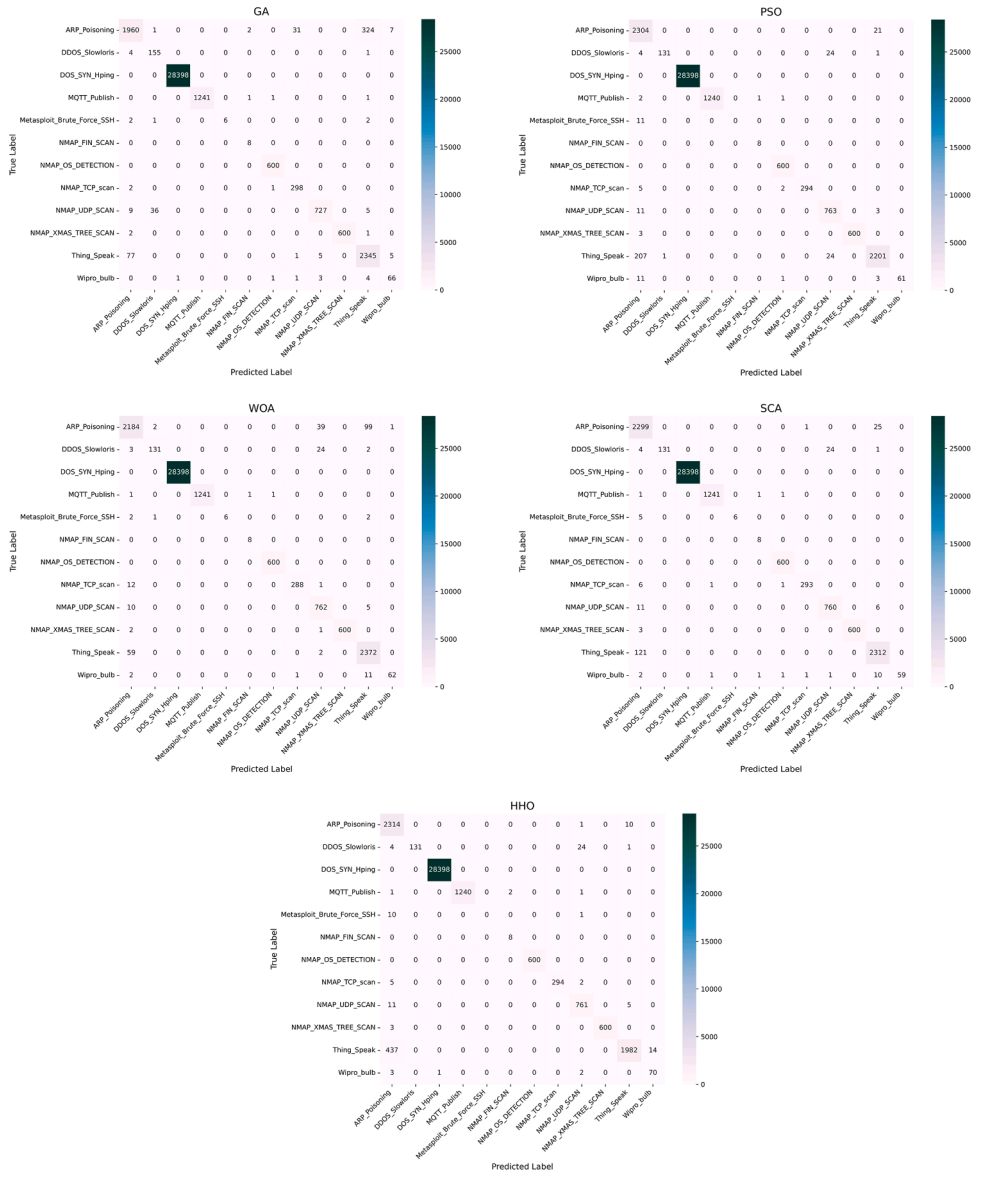


Fig. 7. The confusion matrix of each model.

Table 6
Results obtained from stratified 5-fold cross-validation.

Metric	GA	PSO	WOA	SCA	HHO
Loss	0.0250	0.0485	0.0207	**0.0197**	0.0359
Accuracy	0.9919	0.9882	0.9937	**0.9940**	0.9916
Recall	0.9911	0.9879	0.9933	**0.9936**	0.9915
Precision	0.9923	0.9885	0.9941	**0.9943**	0.9921
F1 Score	0.9917	0.9882	0.9937	**0.9940**	0.9918

feature refinement with automated deep learning configuration, enhancing structural adaptability to complex IoT traffic patterns. The substantial reduction in feature dimensionality (>70%) decreases the input size of the DL model, thereby lowering memory usage and forward-pass computational cost during inference. This reduction contributes to decreased latency and improved throughput, both of which are critical for real-time IoT intrusion detection scenarios. It is important to distinguish between offline optimization cost and online inference cost. While the integration of RF filtering and multiple metaheuristic optimizers increases computational requirements during the model design and training phase, this overhead is incurred only during offline optimization or periodic retraining. Once deployed, the IDS operates using a compact, optimized deep learning architecture without requiring metaheuristic execution, making real-time inference computationally lightweight.

Despite these strengths, several limitations should be acknowledged. The use of five metaheuristic optimizers increases training time and overall computational complexity during the offline optimization stage. Although this improves robustness in feature selection and architecture search, it may require substantial computational resources, particularly for large-scale datasets. Additionally, different optimizers may converge toward partially overlapping feature subsets, potentially reducing diversity gains while still incurring computational cost.

The evaluation is currently limited to the RT-IoT 2022 dataset; therefore, performance may vary when exposed to different traffic distributions or previously unseen attack types. Continuous monitoring, drift detection, and periodic retraining would be necessary in real-world deployments. Furthermore, the feature subset determined during offline training remains static and may not immediately adapt to evolving IoT traffic patterns without incremental learning mechanisms. Although dimensionality reduction lowers inference complexity, the initial RF-

Table 7
Comparative performance of IoT intrusion detection methods in the existing literature.

Studies	Dataset	Class	Method	Performance
Saba et al. [24]	BoT-IoT, NID	2-class, 4-class	CNN	Acc = 99.51%, 95.55%
Kasongo [25]	NSL-KDD, UNSW-NB15	2-class, 5-class; 2-class, 10-class	XGBoost-RNN	Acc = 88.13%, 86.93%; 87.07%, 78.40%
Turukmane and Devendiran [27]	CSECIC-IDS 2018, UNSW-NB15	8-class, 10-class	M-MultiSVM	Acc = 99.89%, 97.54%
Nandanwar and Katarya [28]	N_BaIoT	10-class	AttackNet	Acc = 99.75%
Nandanwar and Katarya [29]	N_BaIoT	10-class	TL-BILSTM	Acc = 99.52%
Al-Ambusaidi et al. [34]	TON-IoT, UNSW-NB15	2-class, 2-class	ML-IDS	Acc = 100%, 92.12%
Hizal et al. [35]	CICIoT2023	12-class	LSTM (two-stage)	Acc = 91.27%
Wang et al. [37]	UNSW-NB15, CIC-IDS2018, CIC-IoT2023	8-class, 15-class, 13-class	ResNeSt + BiGRU	Acc = 91.08%, 94.55%, 97.47%
Hanafi et al. [38]	NSL-KDD, CICIDS2017	5-class, 7-class	IBGJO-LSTM	Acc = 98.93%, 99.75%
Sharma et al. [39]	NSL-KDD, UNSW-NB15	5-class, 5-class	2D-CNN	Acc = 99.4%, 81%
Kharoubi et al. [48]	CICIoT2023, Edge-IIoTset, CICIoMT2024	2-class, 8-class; 2-class, 6-class; 2-class, 6-class	CNN	Acc = 99.42%, 99.17%; 100%, 96.50%; 99.78%, 99.67%
Ogunseyi and Thiyagarajan [49]	NF-BoT-IoT-v2, IoTID20	5-class, 5-class	LSTM	Acc = 98.42%, 89.54%
Our paper	RT-IoT 2022	12-class	Two-stage FS + SCA-DL	Acc = 99.38%

based filtering and multi-optimizer search may present deployment challenges in environments without sufficient centralized computational resources. While the framework is structurally designed to support generalization, explicit validation on unseen attack categories or cross-dataset scenarios remains an important direction for future investigation.

5. Conclusion and future work

The IoT has become an indispensable component of modern digital ecosystems. It enables objects in our daily lives to communicate and exchange data over the internet. However, its rapid and widespread adoption has introduced critical security challenges, as IoT devices often operate with limited protection against cyber threats. To address this issue, this study proposes a novel, fully automated, and scalable framework for intrusion detection in IoT systems. The proposed framework integrates metaheuristic-driven FS and DL model optimization. The key contributions and critical findings of this research are summarized as follows:

- **Two-Stage Feature Selection**: A hybrid FS framework was designed, beginning with RF-based importance ranking to eliminate redundant features. Subsequently, five metaheuristic algorithms (GA, PSO, WOA, SCA, HHO) were independently applied to further refine the feature subset. This approach led to a reduction in feature space by over 70% while preserving the most discriminative features for classification.
- **Metaheuristic-Based DL Optimization**: Each metaheuristic algorithm was used to automatically design and tune the DL model architecture, including the number and type of layers, activation functions, dropout strategies, and learning rates. This automated approach ensures that models are tailored specifically to the characteristics of the IoT dataset, avoiding the need for manual intervention.
- **Superior Model Performance**: Experimental results on the RT-IoT 2022 dataset demonstrated that the SCA-optimized model significantly outperformed the others across all evaluation metrics. It achieved the highest accuracy of 99.38%, the lowest loss of 2.63%, and maintained exceptional generalization with an F1 score of 99.40% in stratified 5-fold cross-validation. These results underline the robustness, precision, and adaptability of the proposed system in detecting a wide variety of network intrusions.
- **Effective Handling of Imbalanced and Complex Data**: The proposed method was able to effectively identify difficult attack classes such as “ARP_Poisoning” and “DDOS_Slowloris”, as well as maintain perfect classification for others like “DOS_SYN_Hping” and “NMAP_OS_DETECTION”. This reflects the model’s capacity to generalize across both frequent and rare attack types in highly imbalanced datasets.

In summary, the study presents a fully automated, high-performance, and interpretable IDS framework that addresses critical security needs in IoT networks. The combination of hybrid FS and architecture-level DL optimization through metaheuristic algorithms represents a novel contribution in this domain. The proposed IDS achieved 99.38% accuracy and 2.63% loss under stratified 5-fold cross-validation, demonstrating stable performance across all evaluation metrics. The low variance observed across folds further supports the generalization capability of the framework within the evaluated dataset. However, real-world IoT environments are dynamic and may introduce previously unseen traffic distributions or novel attack strategies.

Future research will therefore focus on enhancing adaptability and deployment readiness. Planned directions include cross-dataset validation, adversarial robustness evaluation, federated learning for decentralized model updates, and incremental learning mechanisms to address concept drift. Additionally, model compression and edge-oriented optimization strategies will be explored to further reduce deployment overhead. The integration of explainable AI techniques will also be prioritized to enhance transparency and trust in mission-critical IoT security applications.

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Muhammed Furkan Gül: Writing – original draft, Visualization, Software, Resources, Methodology, Funding acquisition, Formal analysis. **Halit Bakır:** Writing – review & editing, Validation, Supervision, Project administration, Investigation, Funding acquisition, Conceptualization.

Declaration of competing interest

The author declares that he has no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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